



MedGuardian AI

Final Review

An AI-Powered Medication Adherence & Drug-Interaction Copilot

Management of AI Products (MBACC ZG583)

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A visibility gap, not a knowledge gap

Patients on three or more chronic prescriptions frequently miss doses or unknowingly combine drugs in harmful ways — a leading, preventable driver of hospital readmission.

1

Physician

Checks interactions once, against what they know, at the moment of prescribing.

2

Pharmacist

Checks again at pickup — still a single point in time.

3

Patient, afterward

Adds an OTC drug, a supplement, a second prescription — invisible to both.

WHY IT MATTERS NOW

3+

chronic prescriptions define the primary polypharmacy patient

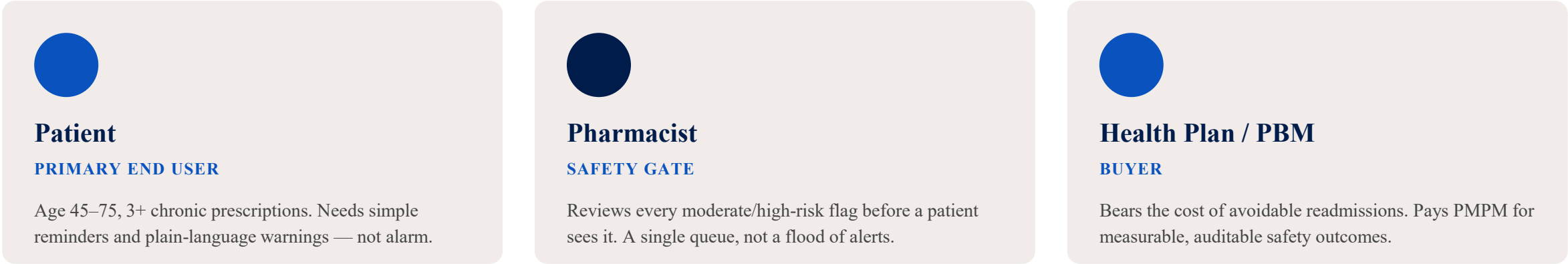
Preventable

missed doses & undetected interactions drive avoidable readmissions

Zero

continuous visibility today — every check is a single point in time

Design for the patient, fund it through the payer



CORE WORKFLOW



Two jobs, one deliberately not a model

Classification — not AI

A deterministic, rules-based lookup flags known interaction pairs against real pharmacological data. No model, no training risk, fully reviewable — because this half must never silently miss a real interaction.

Generation — where AI earns its place

Turning a structured risk flag into calm, plain language a worried patient can actually act on. A rules engine can say two drugs interact; it can't explain what that means for you.

Buildable now, not a research bet:

DrugBank

RxNorm (NLM)

openFDA

All free, structured, and API-accessible today — an engineering effort, not a data-acquisition project.

What ships first, on purpose

MedGuardian AI is a medication adherence and drug-interaction copilot: capture a medication, check it against a real FDA-grounded reference, flag anything risky for pharmacist review, and track adherence daily — designed to embed inside an existing pharmacy, insurer, or hospital app.

Feature	Priority
Barcode/photo medication capture	P0 — MVP
Interaction alert (rules-based + pharmacist review)	P0 — MVP
Daily adherence check-in	P0 — MVP
GenAI plain-language explanation	P1 — Fast follow
Caregiver dashboard	P1
Personalized adherence-risk prediction	P2 — Scale
EHR / pharmacy system integration	P2 — Scale

MVP DEFINITION

Successful if it can reliably flag known interactions with zero false negatives, hold every moderate/high-risk case for pharmacist review, and log adherence without needing any live AI call to function.

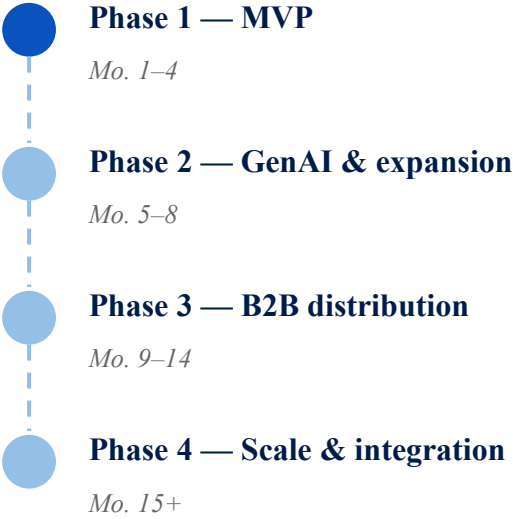
Single condition · One language · Zero AI dependency in the safety-critical path

Build vs. buy, and the roadmap it drives

BUILD vs. BUY vs. PARTNER/API

Drug data	Buy / API	DrugBank, RxNorm, openFDA — a solved problem
Risk-scoring logic	Build	The product's core, differentiated IP
Explanation layer	Buy model + Build guardrails	General LLM API, in-house safety prompt
Distribution	Partner	Health plans, PBMs, pharmacy chains

ROADMAP

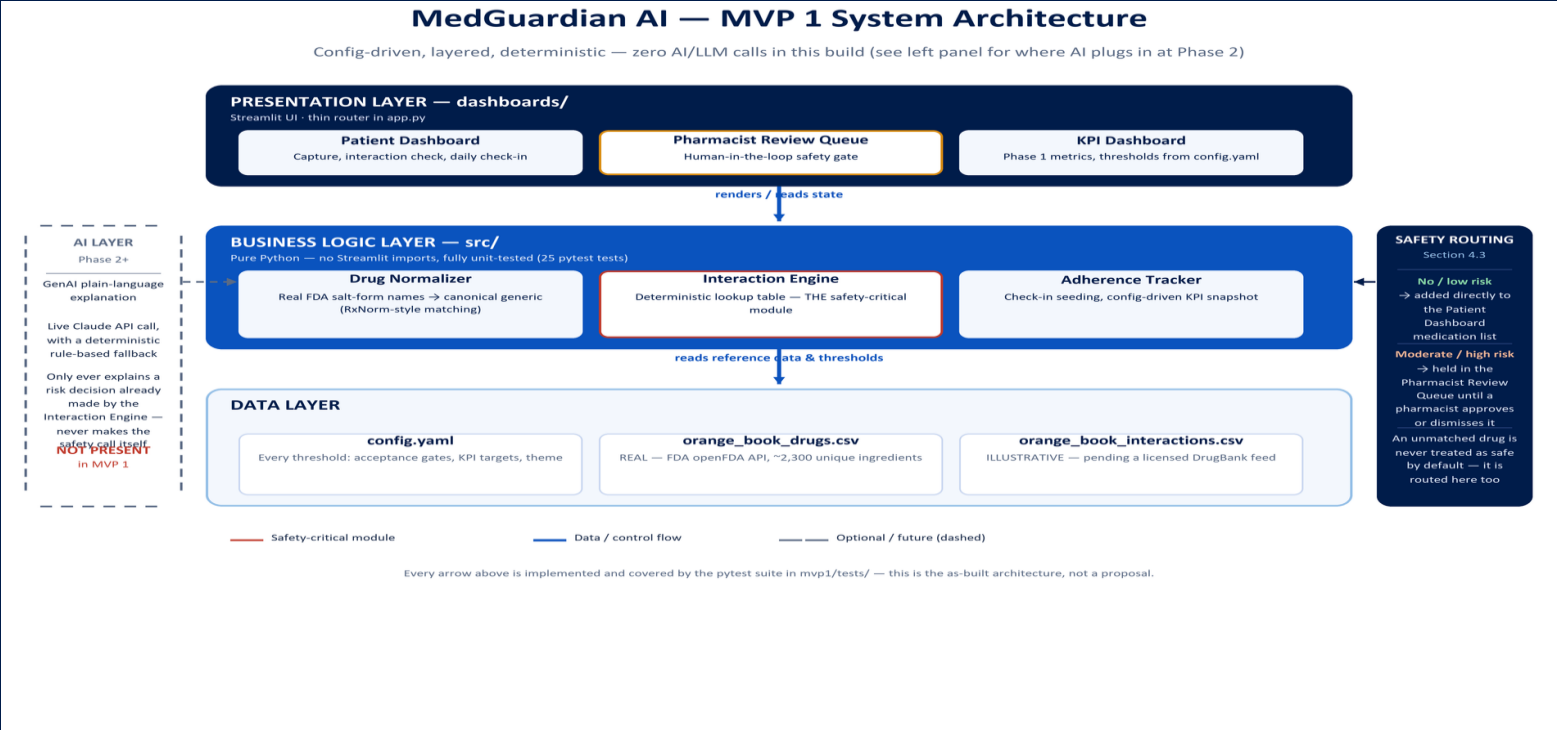


FIRST WEDGE

Single condition (Type 2 diabetes with cardiometabolic comorbidities), one language, a focused drug reference set — prove the pharmacist-agreement gate before expanding.

6 — PROTOTYPE WALKTHROUGH

MVP 1 — built, tested, and running



47

passing automated tests

Zero

live-AI calls in the safety-critical path

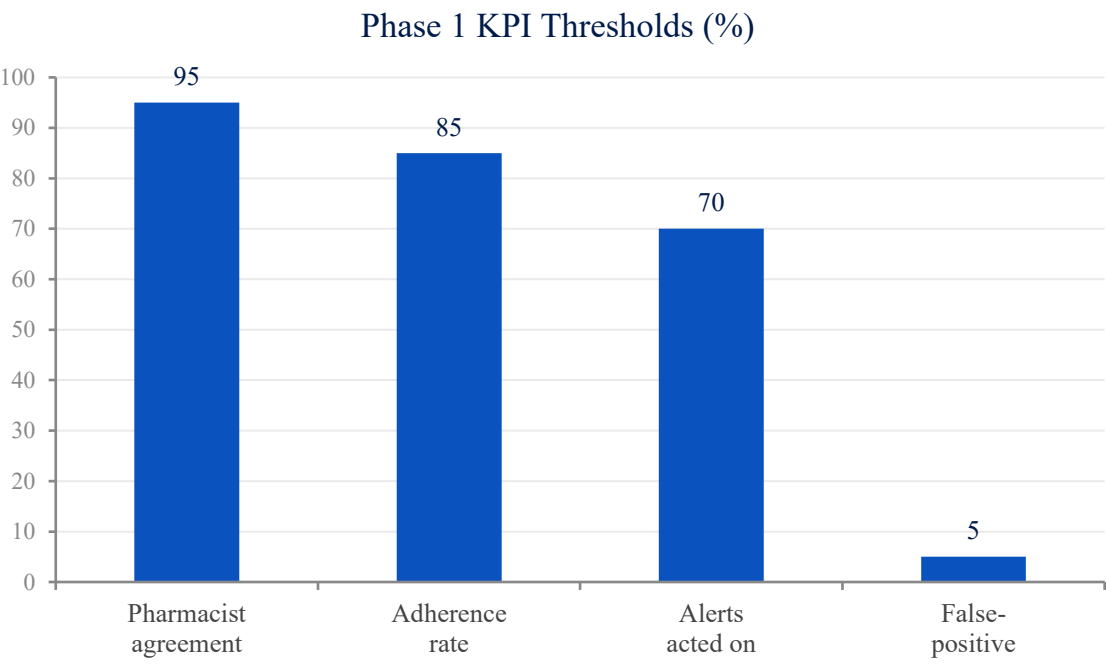
Real

FDA Orange Book data — not a mock list

3

capture / check / review flows fully working

What proves it, what pays for it



MONETIZATION

- **B2B2C — PMPM licensing**
Primary — health plans / PBMs
- **DTC freemium**
Secondary — premium tier
- **API licensing**
Expansion — pharmacy / EHR vendors

LAUNCH BLOCKER

If pharmacist agreement cannot be sustained $\geq 95\%$, or false-positive rate kept under 5%, the product does not move beyond a narrow pilot — regardless of engineering readiness.

8 — FINAL RECOMMENDATION

PILOT

The rules-based engine, the pharmacist-review workflow, and a tested MVP 1 prototype are already built. The next step is a scoped, time-boxed pilot — measured against $\geq 95\%$ pharmacist agreement and a false-positive rate under 5% — not more prototyping.

Top-3 driver

of preventable hospital readmission

Buildable today

on free, structured pharmacological data

Human-reviewed

from day one — automation is earned





Thank You

Questions & Discussion